

Advanced (Habanero)

Q1. Rationalize the denominator of $\frac{1}{\sqrt{2}}$

- (A) $\frac{\sqrt{2}}{2}$
- (B) $\frac{1}{2\sqrt{2}}$
- (C) $\frac{2}{\sqrt{2}}$
- (D) $\frac{\sqrt{2}}{4}$
- (E) $\frac{1}{4\sqrt{2}}$

Q2. Simplify $\frac{1}{\sqrt{5}+\sqrt{3}}$

- (A) $\frac{\sqrt{5}-\sqrt{3}}{2}$
- (B) $\frac{\sqrt{5}+\sqrt{3}}{2}$
- (C) $\frac{\sqrt{15}-3}{2}$
- (D) $\frac{\sqrt{15}+3}{2}$
- (E) $\frac{\sqrt{15}-1}{2}$

Q3. Rationalize the denominator of $\frac{2}{\sqrt{6}-\sqrt{2}}$

- (A) $\frac{2(\sqrt{6}+\sqrt{2})}{4}$
- (B) $\frac{2(\sqrt{6}-\sqrt{2})}{4}$
- (C) $\frac{\sqrt{6}+\sqrt{2}}{2}$
- (D) $\frac{\sqrt{6}-\sqrt{2}}{2}$
- (E) $\frac{2(\sqrt{6}+\sqrt{2})}{2}$

Q4. Simplify $\frac{1}{\sqrt{7}-\sqrt{3}}$

- (A) $\frac{\sqrt{7}+\sqrt{3}}{4}$
- (B) $\frac{\sqrt{7}-\sqrt{3}}{4}$
- (C) $\frac{\sqrt{21}+3}{4}$
- (D) $\frac{\sqrt{21}-3}{4}$
- (E) $\frac{\sqrt{21}+1}{4}$

Q5. Rationalize the denominator of $\frac{3}{\sqrt{10}+\sqrt{2}}$

- (A) $\frac{3(\sqrt{10}-\sqrt{2})}{8}$
- (B) $\frac{3(\sqrt{10}+\sqrt{2})}{8}$
- (C) $\frac{\sqrt{10}-\sqrt{2}}{3}$
- (D) $\frac{\sqrt{10}+\sqrt{2}}{3}$
- (E) $\frac{3(\sqrt{10}-\sqrt{2})}{3}$

Q6. Simplify $\frac{1}{\sqrt{9}+\sqrt{4}}$

- (A) $\frac{1}{5}$
- (B) $\frac{1}{3}$
- (C) $\frac{1}{7}$
- (D) $\frac{1}{13}$
- (E) $\frac{1}{17}$

Q7. Rationalize the denominator of $\frac{2}{\sqrt{3}+1}$

- (A) $\frac{2(\sqrt{3}-1)}{2}$
- (B) $\frac{2(\sqrt{3}+1)}{2}$
- (C) $\frac{\sqrt{3}-1}{2}$
- (D) $\frac{\sqrt{3}+1}{2}$
- (E) $\frac{2(\sqrt{3}-1)}{3}$

Q8. Simplify $\frac{1}{\sqrt{8}-\sqrt{2}}$

- (A) $\frac{\sqrt{8}+\sqrt{2}}{6}$
- (B) $\frac{\sqrt{8}-\sqrt{2}}{6}$
- (C) $\frac{\sqrt{8}+\sqrt{2}}{2}$
- (D) $\frac{\sqrt{8}-\sqrt{2}}{2}$
- (E) $\frac{\sqrt{8}+\sqrt{2}}{4}$

Open Ended Questions (FRQ)

Q9. Rationalize the denominator of $\frac{1}{\sqrt{x}+\sqrt{y}}$

Q10. Simplify $\frac{2}{\sqrt{a}-\sqrt{b}}$ and rationalize the denominator

Chef's Tips

- Encourage students to show all work and simplify answers
- Remind students to check for extraneous solutions
- Have students use online resources to check their answers